

GitHub - sinsinology/CVE-2022-26503 · GitHub

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- Repository commit: 5f031eaf0bed6adcdfaa3079a63694b69db3d170
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Content

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Repository reading copy: selected documentation at the recorded commit. Source code is never checked out, built or run.

- Repository: <https://github.com/sinsinology/CVE-2022-26503>
- Commit: 5f031eaf0bed6adcdfaa3079a63694b69db3d170
- Documents preserved: 1

README.md

Blob 5c867f72ec60, 1295 bytes, at commit 5f031eaf0bed.

CVE-2022-26503

Summary:

Vulnerability (CVE-2022-26503) in Veeam Agent for Microsoft Windows allows local privilege escalation. An attacker who successfully exploited this vulnerability could run arbitrary code with LOCAL SYSTEM privileges.

Shout out to @ultrayoba

The Patch:

The implemented patch shows blood trail of Deserialization:

```
BinaryServerFormatterSinkProvider binaryServerFormatterSinkProvider = new BinaryServerFormatterSinkProvider()
{
    TypeFilterLevel = TypeFilterLevel.Full
};
IDictionary strs = new Dictionary<string, string>();
strs["name"] = channelName;
strs["port"] = port.ToString();
strs["authenticationMode"] = options.AuthenticationMode;

IServerChannelSinkProvider sinkProvider = CSrvTcpChannelRegistration.GetSinkProvider();

IDictionary strs = new Dictionary<string, string>();
strs["name"] = channelName;
strs["port"] = port.ToString();
strs["authenticationMode"] = options.AuthenticationMode;
```

Veeam official KB mentions:

Veeam Agent for Microsoft Windows uses Microsoft .NET data serialization mechanisms. A local user may send malicious code to the network port opened by Veeam Agent for Windows Service (TCP 9395 by default), which will not be deserialized properly.

Analysis:

Reviewing process behind the specified port results in finding Veeam.Endpoint.Service.exe

```
C:\Users\SinSinology>netstat -ano | findstr 9395
TCP        0.0.0.0:9395        0.0.0.0:0           LISTENING        7272
TCP        127.0.0.1:9395      127.0.0.1:49914     ESTABLISHED      7272
TCP        127.0.0.1:49914     127.0.0.1:9395     ESTABLISHED      3388
TCP        127.0.0.1:49915     127.0.0.1:9395     TIME_WAIT        0

C:\Users\SinSinology>tasklist | findstr 7272
Veeam.Endpoint.Service.exe      7272 Services                0      51,468 K

C:\Users\SinSinology>
```

Reviewing Veeam.Endpoint.Service.exe indicates registration of VeeamService for .NET Remoting

```
C:\EndPointServiceStub @0200001F
Base Type and Interfaces
Derived Types
C:\EndPointServiceStub: void @06000175
CreateSvcBackupSessionObj(IpServiceImpl) : IService
InitialSvcLifetimeService() : object @06000178
StartLeasing(IpServiceImpl) : void @06000177
StopLeasing() : void @06000178
Veeam.Backup.Interaction.Management.VeeamBack
Veeam.Backup.Interaction.Management.VeeamBack
m_instanceName : CVbSvcName @0400009E
s_serviceImpl : IPServiceImpl @0400009D

25 // Token: 0x06000177 RID: 375 RVA: 0x0000826D File Offset: 0x0000646D
26 internal static void StartLeasing(IPServiceImpl serviceImpl)
27 {
28     CEndPointServiceStub.s_serviceImpl = serviceImpl;
29     SVeeamBackupService.Instance.SetSvcSessionObj(CEndPointServiceStub.CreateSvcBackupSessionObj
30         (CEndPointServiceStub.s_serviceImpl));
31     RemotingConfiguration.RegisterWellKnownServiceType(typeof(CEndPointServiceStub), "VeeamService",
32         WellKnownObjectMode.Singleton);
33 }
```

Processes communicating with the registered channel gives out information about Veeam.Endpoint.Tray.exe showing this channel gets used by Tray process

```
C:\Users\SinSinology>netstat -ano | findstr 9395
TCP    0.0.0.0:9395          0.0.0.0:0             LISTENING        7272
TCP    127.0.0.1:9395       127.0.0.1:49914       ESTABLISHED      7272
TCP    127.0.0.1:49914     127.0.0.1:9395       ESTABLISHED      3388
TCP    127.0.0.1:49915     127.0.0.1:9395       TIME_WAIT        0

C:\Users\SinSinology>tasklist | findstr 7272
Veeam.EndPoint.Service.exe    7272 Services                0    51,468 K

C:\Users\SinSinology>tasklist | findstr 3388
Veeam.EndPoint.Tray.exe       3388 Console                    1    74,152 K

C:\Users\SinSinology>
```

Loaded modules by the Tray indicate Veeam.Common.Remoting.dll

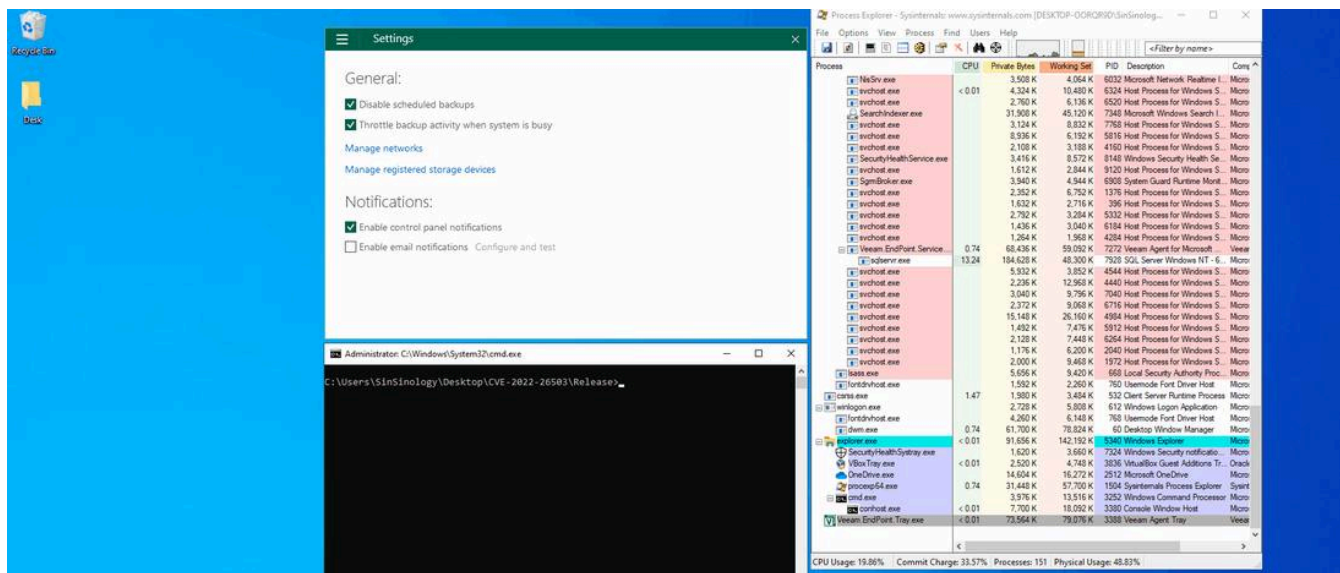
 Veeam.EndPoint.Tray.exe	< 0.01	73,356 K	75,240 K	3388 Veeam Agent Tray	Veeam Software Group G...
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Name	Description	Company Name	Path
Veeam.Backup.Interaction.Management.dll	Veeam.Backup.Interaction.Manag...	Veeam Software Group G...	C:\Program Files\Veeam\Endpoint Backup\Veeam.Backup....
Veeam.Backup.Model.dll	Veeam.Backup.Model	Veeam Software Group G...	C:\Program Files\Veeam\Endpoint Backup\Veeam.Backup....
Veeam.Backup.ServiceConnection.dll	Veeam.Backup.ServiceConnection	Veeam Software Group G...	C:\Program Files\Veeam\Endpoint Backup\Veeam.Backup....
Veeam.Common.Remoting.dll	Veeam.Backup.Common.Remoting	Veeam Software Group G...	C:\Program Files\Veeam\Endpoint Backup\Veeam.Common...
Veeam.EndPoint.Common.dll	Veeam.EndPoint.Common	Veeam Software Group G...	C:\Program Files\Veeam\Endpoint Backup\Veeam.EndPoint...
Veeam.EndPoint.Core.dll	Veeam.EndPoint.Core	Veeam Software Group G...	C:\Program Files\Veeam\Endpoint Backup\Veeam.EndPoint...
Veeam.EndPoint.Presentation.dll	Veeam.EndPoint.Presentation	Veeam Software Group G...	C:\Program Files\Veeam\Endpoint Backup\Veeam.EndPoint...
Veeam.EndPoint.Tray.exe	Veeam Agent Tray	Veeam Software Group G...	C:\Program Files\Veeam\Endpoint Backup\Veeam.EndPoint...

Use of TcpClientChannel with enabled Secure

```
this.m_channelName = channelName;
if (ChannelServices.GetChannel(channelName) != null)
{
    return;
}
Log.Message("Registering TCP client channel [{0}]", new object[]
{
    channelName
});
routerSinkProvider.Next = new CBinaryClientFormatterSinkProvider();
Dictionary<string, string> dictionary = new Dictionary<string, string>();
dictionary["name"] = channelName;
dictionary["tokenImpersonationLevel"] = "Identification";
dictionary["secure"] = "true";
dictionary["timeout"] = "900000";
if (options != null)
{
    options.ApplyChannelProperties(dictionary);
}
ChannelServices.RegisterChannel(new TcpClientChannel(dictionary, routerSinkProvider)
{
    IsSecured = true
}, true);
}
```

Exploit:



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